A CASE-STUDY BASED REPORT

ON

Wine Quality Prediction from using Ridge, Lasso and Elastic Net Regression

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

A.Y.: 2026-27

by

P. Bhavana Manaswini **roll 2520030154**

Kassireddy Srihitha **roll 2520030153**

Under the Esteemed guidance of

Mr. Shaik Asif

Asst.Professor, Dept. of CSE

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

INDEX

Abstract **3**

Introduction **3**

Background 3
Motivation for the Project 3
Overview of the Work Carried Out 3
Scope and Responsible-Use Statement 4

Problem Statement **4**

Case-Study Scenario 4
Central Case-Study Question 4
Formal Problem Definition 4
Objectives that Guide the Study 4
Assumptions and Limitations 5

Dataset and Exploratory Data Analysis **5**

About the Dataset 5
Purpose of EDA 5
Preparation Steps Actually Performed 5
EDA Exploration - Implementation 6
Summary of EDA Findings 6

Algorithm Selection and Implementation **6**

Conclusion **8**

Abstract

Assessing the quality of red wine has traditionally depended on sensory evaluation by trained tasters, a process that is subjective, slow and difficult to scale. This case study investigates whether the quality score of a red wine sample can be estimated directly from its laboratory-measured physicochemical properties. Using the WineQT dataset of 1,143 samples described by 11 physicochemical features, the problem is formulated as a supervised regression task in which the numerical quality score is the target variable.

Three regularized linear regression models are implemented and compared: Ridge Regression (L2 penalty), Lasso Regression (L1 penalty) and Elastic Net Regression (combined L1 and L2 penalty). The data is split 80/20 into training and testing sets with random_state = 42, features are standardized with StandardScaler, and hyperparameters are tuned using 5-fold cross-validated GridSearchCV. Model performance is measured with MAE, MSE, RMSE and R^2. The study demonstrates how a standard machine-learning workflow, from exploratory data analysis to model comparison, can support objective and repeatable wine-quality assessment.

Introduction

Background

Wine quality is influenced by many chemical characteristics such as acidity, sugar, sulphur dioxide levels, density, pH, sulphates and alcohol content. Producers routinely measure these properties in the laboratory, yet the final quality rating is still commonly assigned by human tasters. Sensory judgement is valuable but varies from person to person and is costly to repeat for every batch. Machine learning offers a way to learn the relationship between measurable chemistry and perceived quality from historical data.

Motivation for the Project

A producer receiving many batches of red wine needs a fast and consistent first-level quality estimate. A data-driven model trained on laboratory measurements can provide such an estimate and reduce the dependence on purely subjective assessment. Physicochemical features of wine are also correlated with one another (for example acidity, citric acid, density and pH), which makes ordinary least-squares regression unstable. Regularized regression methods are therefore a natural choice, since they control coefficient size and, in the case of Lasso and Elastic Net, can perform feature selection.

Overview of the Work Carried Out

The work covers the complete regression workflow: loading and preprocessing the WineQT dataset, exploratory data analysis (EDA), separating features from the target and removing the identifier column, splitting the data 80/20, scaling the features, training Ridge, Lasso and Elastic Net models with 5-fold cross-validated hyperparameter tuning, predicting quality scores on unseen data, and comparing the models using MAE, MSE, RMSE and R^2.

Scope and Responsible-Use Statement

The study is limited to red wine samples in the WineQT dataset and to linear regularized regression models. The predicted score is intended as a decision-support aid for quality assessment and does not replace trained tasters or regulatory testing. The dataset contains only physicochemical measurements and a quality score; it holds no personal information. Results should not be generalized to white wines, other wine regions or measurement setups without re-validation.

Problem Statement

Case-Study Scenario

A wine producer receives different batches of red wine that must be assessed for quality. Traditional assessment relies on sensory evaluation by trained tasters, which can be subjective and time-consuming. The producer has laboratory measurements describing the physicochemical properties of each wine sample and wants to use these measurements to support quality assessment.

Central Case-Study Question

How accurately can the quality score of a red wine sample be predicted from its physicochemical measurements, and which regularization strategy (Ridge, Lasso or Elastic Net) gives the best predictive performance?

Formal Problem Definition

Let the dataset be $D = \{(x_i, y_i)\}$, $i = 1, \ldots, n$, where x_i is a vector of 11 physicochemical features and y_i is the quality score of sample i (observed range 3 to 8). The goal is to learn a function f that maps x_i to a predicted score $\hat{y}_i = f(x_i)$ so that the prediction error on unseen samples is minimized. Because the target is numeric, this is a supervised regression problem. The three models differ in the penalty added to the squared-error loss:

- Ridge: minimize $\|y - Xw\|^2 + \alpha\|w\|_2^2$ (L2 penalty shrinks all coefficients).
- Lasso: minimize $\|y - Xw\|^2 + \alpha\|w\|_1$ (L1 penalty can set some coefficients exactly to zero).
- Elastic Net: minimize $\|y - Xw\|^2 + \alpha[\rho\|w\|_1 + (1 - \rho)\|w\|_2^2]$ (mixture of both, with mixing ratio ρ).

Objectives that Guide the Study

1. Explore the WineQT dataset and understand the distribution of the quality score and the relationships among physicochemical features.
2. Prepare the data by removing the identifier, splitting it 80/20 and standardizing the features.
3. Build Ridge, Lasso and Elastic Net regression models and tune their hyperparameters with 5-fold cross-validated GridSearchCV.
4. Evaluate each model on the test set using MAE, MSE, RMSE and R^2.
5. Compare the three regularization strategies and identify the most suitable model for estimating wine quality.

Assumptions and Limitations

- The quality score is treated as a continuous numeric target, although it is originally an integer rating from sensory evaluation.
- The relationship between chemistry and quality is assumed to be approximately linear after scaling; non-linear effects and feature interactions are not modelled.
- Quality scores are concentrated in the middle of the range, so extreme ratings (very poor or excellent wines) are harder to predict.
- The sample is limited to 1,143 red wine records, and sensory scores themselves carry some subjectivity and noise.

Dataset and Exploratory Data Analysis

About the Dataset

The study uses the WineQT.csv dataset, which contains physicochemical test results and sensory quality ratings for red wine samples.

Dataset Property	Details
Records	1,143
Columns	13
Predictor variables	11 physicochemical features
Target variable	quality (numeric score, range 3–8)
Identifier	Id (excluded from training)

The 11 input features are: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and alcohol. The Id column is only a record identifier and carries no predictive information, so it is removed before modelling.

Purpose of EDA

EDA is carried out to understand the structure and quality of the data before any model is built. It helps to check for missing or inconsistent values, to examine how the quality score is distributed, to detect outliers and skewed features, and to study the correlation between features and with the target. These observations justify the choice of feature scaling and of regularized models.

Preparation Steps Actually Performed

1. Loaded WineQT.csv and inspected its shape, data types and summary statistics.
2. Separated the 11 predictor features from the target variable (quality).
3. Removed the Id column because it is only an identifier.
4. Split the data into 80% training and 20% testing sets using random_state = 42 for reproducibility.

5. Applied StandardScaler so that all features have zero mean and unit variance, so that the regularization penalty treats every feature fairly.

EDA Exploration - Implementation

The exploratory analysis was implemented in Python and covered the following views of the data:

- Descriptive statistics and a missing-value check for all columns.
- Distribution of the target variable (count of samples per quality score).
- Histograms of each physicochemical feature to study skewness and spread.
- Box plots to identify outliers in features such as residual sugar, chlorides and sulphur dioxide.
- A correlation heatmap to study relationships among features and between each feature and quality.

[Insert Figure 1: Distribution of wine quality scores, from the EDA notebook]

[Insert Figure 2: Correlation heatmap of the physicochemical features and quality, from the EDA notebook]

Summary of EDA Findings

- Most wines receive middle quality scores, while very low and very high scores are rare, so the target is imbalanced towards the centre of the range.
- Alcohol shows a positive association with quality, whereas volatile acidity shows a negative association.
- Several features are correlated with each other (for example fixed acidity, citric acid, density and pH), which supports the use of regularized regression to control multicollinearity.
- Some features are right-skewed and contain outliers, and the features are on very different scales, which makes standardization necessary.

Algorithm Selection and Implementation

Why Regularized Regression

Plain linear regression becomes unreliable when predictors are correlated, because coefficient estimates grow large and unstable. Regularization adds a penalty on coefficient size, which improves generalization to unseen samples. Three variants are compared so that the effect of each penalty type can be observed on the same data.

Model	Penalty	Behaviour
Ridge Regression	L2	Shrinks coefficient magnitudes and helps control multicollinearity; keeps all features.
Lasso Regression	L1	Can shrink some coefficients to exactly zero, giving feature-selection behaviour.
ElasticNet Regression	L1 + L2	Balances feature selection and coefficient stability; useful when features are correlated.

Implementation Workflow

The case-study flow is: Wine Quality Dataset → Data Preprocessing → Exploratory Data Analysis → Feature / Target Separation → Remove Id → 80/20 Train-Test Split → Standard Scaling → Ridge | Lasso | Elastic Net → Prediction → Model Evaluation (MAE, MSE, RMSE, R^2) → Model Comparison.

1. The scaler is fitted on the training data only and then applied to the test data, to avoid information leakage.
2. Each model is tuned with 5-fold cross-validated GridSearchCV over its regularization hyperparameters (alpha for all three models, and the L1 ratio for Elastic Net).
3. The best estimator of each search is used to predict quality scores on the held-out test set.
4. Predictions are evaluated with the four regression metrics below.

Evaluation Metrics

- MAE (Mean Absolute Error): average absolute difference between actual and predicted quality.
- MSE (Mean Squared Error): average squared difference; penalizes large errors more heavily.
- RMSE (Root Mean Squared Error): square root of MSE, expressed in the same units as the quality score.
- R^2 (coefficient of determination): proportion of variance in quality explained by the model.

Model Comparison

Lower MAE, MSE and RMSE and a higher R^2 indicate a better model. The test-set results of the three tuned models are summarized below.

Model	Train R^2	5-Fold CV R^2 (mean ± SD)	Test R^2	Train RMSE	Test RMSE
Ridge	0.381	0.351 ± 0.038	0.327	0.645	0.612
Lasso	0.379	0.353 ± 0.038	0.335	0.646	0.608
Elastic Net	**0.378**	**0.353 ± 0.036**	**0.337**	**0.647**	**0.608**

Expected output of the system: for a given wine sample, the physicochemical measurements are passed to the trained regression model, which returns a predicted quality score (Physicochemical Measurements → Regression Model → Predicted Quality Score).

Conclusion

This case study showed how the physicochemical properties of red wine can be used in a machine-learning regression workflow to estimate wine quality. Following exploratory analysis, preprocessing and scaling, Ridge, Lasso and Elastic Net regression were trained with cross-validated tuning and compared using MAE, MSE, RMSE and R^2. The study illustrates how different regularization strategies handle correlated features and how they can be compared fairly on the same train-test split.

The approach offers a fast, objective and repeatable first-level estimate of wine quality that can complement sensory evaluation. Future work could include non-linear models such as random forests or gradient boosting, treating the problem as classification, handling class imbalance, and validating the model on additional wine datasets.